

Phase 2: Boiling Water to Make Lightning

Deconstructing the Steam Power Plant (The Rankine Cycle)

The Virtual Professor

The Professor's Intuition: Why Water?

To turn a massive generator at 3000 RPM, we need to hit its blades with an incredible physical force. We cannot just blow wind at it. We need something that expands violently. When liquid water turns into steam, it expands to **1600 times its original volume**. If you trap that water inside a steel box (a boiler) and heat it, that expansion has nowhere to go. The pressure builds to over 150 atmospheres. When you finally open a valve, that steam shoots out at supersonic speeds. *That* is what spins the turbine. A thermal power plant is just a highly controlled pipe bomb.

1 The Anatomy of the Beast: The Four Flow Circuits

A thermal power plant is incredibly complex, but it can be easily understood by breaking it down into four distinct, interconnected circulatory systems (just like the human body).

1. **Coal and Ash Circuit (The Digestive System):** Coal goes in, gets burned, and the waste (ash) is removed.
2. **Air and Flue Gas Circuit (The Respiratory System):** Fresh air is sucked in to burn the coal, and toxic hot gases (flue gas) are exhaled out the chimney.
3. **Cooling Water Circuit (The Perspiration System):** Cold water from a river cools the exhaust steam back into liquid so it can be reused.
4. **Feed Water and Steam Circuit (The Bloodstream):** This is the most important loop. Water becomes high-pressure steam, does work, and turns back into water.

1.1 Exam Focus: The Feed Water and Steam Circuit

(Frequently asked in JU Exams: 2024 Part I Q1, 2022 Part I Q2a)

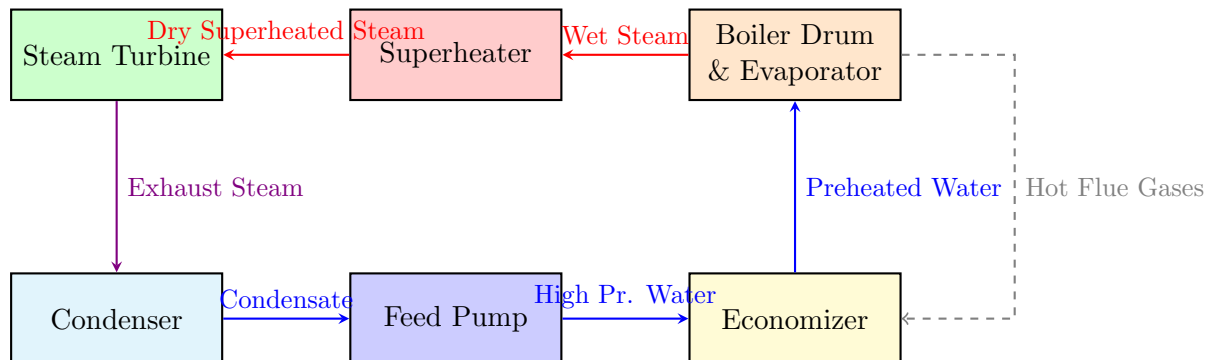


Figure 1: The Closed-Loop Rankine Cycle (Feed Water and Steam Circuit)

2 The Efficiency Boosters (Heat Recovery)

(Asked in 2022 Part I Q2c)

When the coal burns, the resulting "Flue Gases" are incredibly hot ($> 1000^{\circ}\text{C}$). If we just let them go out the chimney, we throw away millions of dollars of heat. So, we place three "toll booths" in the exhaust pipe to steal that heat back before the gas escapes:

1. **The Superheater:** The boiler produces "wet steam" (steam with tiny water droplets). If these water droplets hit the turbine blades at supersonic speeds, they act like bullets and destroy the metal. The Superheater steals heat from the flue gas to heat the steam *above* its boiling point, making it 100% dry and safe.
2. **The Economizer:** It steals heat from the flue gas to pre-heat the liquid feed water *before* it enters the boiler. If the water is already hot, the boiler uses less coal.
3. **The Air Preheater:** It steals the last bit of usable heat from the flue gas to warm up the fresh atmospheric air *before* it enters the furnace. Hot air makes the coal burn faster and hotter.

3 The Lungs: Draught Systems

To burn hundreds of tons of coal an hour, a fire needs millions of cubic meters of oxygen. A normal chimney (Natural Draught) cannot pull in air fast enough. We must use massive industrial fans (Artificial Draught).

Exam Focus: Why Balanced Draught? (2024, 2022 Exams)

- **Forced Draught (FD) Only:** A fan pushes air *into* the furnace. **Problem:** The furnace is at positive pressure. If you open an inspection door, toxic gas, ash, and flames will blow out into your face! (Highly dangerous).
- **Induced Draught (ID) Only:** A fan at the chimney sucks air *out* of the furnace. **Problem:** The furnace is at a strong negative pressure (vacuum). Cold air from the outside will leak into the boiler through every tiny crack, chilling the fire and ruining efficiency.
- **The Solution (Balanced Draught):** We use **BOTH**. The FD fan pushes air in, and the ID fan pulls gases out. By balancing their speeds, we keep the furnace at a *very slight vacuum*. This is safe (flames don't blow out) and efficient (prevents massive cold air leaks).

4 The Digestive System: Pulverized Coal & Ash

(Asked in 2025 Part I Q1a, Q1b)

Why Pulverize the Fuel? Try lighting a solid log of wood with a match. It takes forever. Now turn that log into sawdust and throw it in the air with a match—it explodes instantly. By grinding coal into a fine face-powder (pulverization), we dramatically increase the **surface area**. When blown into the furnace, it burns instantly in mid-air like a gas, yielding huge thermal efficiency and rapid response to load changes.

The Fly Ash Problem: Because the coal is burned as a fine dust, the resulting ash is also a microscopic dust that floats in the hot exhaust gas ("Fly Ash"). If released, it would blanket cities in toxic dust.

The Solution: Electrostatic Precipitator (ESP). The flue gas passes between metal plates charged to 50,000 Volts. The intense electric field (corona discharge) gives the ash particles a negative charge. They act like magnets and stick to grounded metal collection plates, leaving the exhaust air clean.

5 The Master Derivation: Stoichiometric Air Calculation

(Appeared in 2024 Exam. Highly conceptual.)

To burn fuel perfectly, we need exact amounts of Oxygen. Let C , H_2 , O_2 , and S be the mass fractions of Carbon, Hydrogen, Oxygen, and Sulfur in 1 kg of fuel.

- **Carbon** ($C + O_2 \rightarrow CO_2$): 12kg of C needs 32kg of O_2 . $\Rightarrow$ 1kg of C needs $\frac{32}{12} = \frac{8}{3}$ kg of O_2 .
- **Hydrogen** ($2H_2 + O_2 \rightarrow 2H_2O$): 4kg of H_2 needs 32kg of O_2 . $\Rightarrow$ 1kg of H_2 needs 8kg of O_2 .
- **Sulfur** ($S + O_2 \rightarrow SO_2$): 32kg of S needs 32kg of O_2 . $\Rightarrow$ 1kg of S needs 1kg of O_2 .

Total Oxygen Required = $\left(\frac{8}{3}C + 8H_2 + S\right)$.

Since the fuel already contains O_2 , we subtract it: **Net Oxygen** = $\left(\frac{8}{3}C + 8H_2 + S - O_2\right)$.

Since normal air is only 23% oxygen by weight, we must multiply the required oxygen by $\frac{100}{23}$ to find the total mass of *air* required:

Theoretical Air Formula

$$\text{Mass of Air} = \frac{100}{23} \left[\frac{8}{3}C + 8 \left(H_2 - \frac{O_2}{8} \right) + S \right] \text{ kg per kg of fuel} \quad (1)$$

6 Semester Application: Plant Efficiency Numericals

(Appeared in 2024, 2022 Exams)

Problem Statement: A 120 MW steam station uses coal with a Calorific Value (CV) of 6400 kcal/kg. Thermal efficiency is 30% and electrical efficiency is 92%. Calculate the coal consumption per hour at full load.

Step 1: Convert Output to Heat Equivalent

Output = 120 MW = 120,000 kW. In one hour, energy = 120,000 kWh.

Using the conversion 1 kWh = 860 kcal,

Heat Output = $120,000 \times 860 = 103,200,000$ kcal/hr.

Step 2: Calculate Overall Efficiency

$\eta_{\text{overall}} = \eta_{\text{thermal}} \times \eta_{\text{electrical}} = 0.30 \times 0.92 = 0.276$.

Step 3: Calculate Required Heat Input

Heat Input = $\frac{\text{Heat Output}}{\eta_{\text{overall}}} = \frac{103,200,000}{0.276} = 373,913,043$ kcal/hr.

Step 4: Calculate Coal Mass

Mass = $\frac{\text{Heat Input}}{\text{Calorific Value}} = \frac{373,913,043}{6400} = 58,423.9$ kg/hr (or 58.4 tonnes/hr).

Your Next Step

You have mastered the fire and the water. But how does that spinning steel shaft safely connect to the grid? **To continue, reply with:**

"Professor, take me to Phase 3. Let's learn about Turbo-Alternators, Hydrogen Cooling, and the mechanical-electrical bridge."